

Class activity: maximum likelihood estimation

Group members:

Instructions: Work with a neighbor on the following activity. I will collect the handout at the end of class, and it will be part of your class participation grade. You will be graded only on effort – it is ok if you don't finish all the questions, or get them all correct.

Maximum likelihood estimation

Suppose we have a random variable Y , which can take on values $y = 1, 2, 3, 4, \dots$ (i.e., any positive integer). The probability of each outcome does not depend on any explanatory variables, and we are told that

$$P(Y = y|\pi) = (1 - \pi)^{y-1}\pi$$

where $0 \leq \pi \leq 1$. So, $P(Y = 1|\pi) = \pi$, $P(Y = 2|\pi) = (1 - \pi)\pi$, etc. And for an arbitrary observation Y_i , we have $P(Y_i|\pi) = (1 - \pi)^{Y_i-1}\pi$.

We don't know π , but we get n independent observations $Y_1, \dots, Y_n$. We will use the method of maximum likelihood to estimate π .

Questions

1. Using independence, the likelihood is defined as

$$L(\pi) = P(Y_1|\pi)P(Y_2|\pi) \cdots P(Y_n|\pi) = \prod_{i=1}^n P(Y_i|\pi)$$

Show that

$$L(\pi) = (1 - \pi)^{\sum_{i=1}^n (Y_i-1)} \pi^n = (1 - \pi)^{\sum_{i=1}^n Y_i - n} \pi^n$$

2. Using calculus (take the log and differentiate the log-likelihood), show that the maximum likelihood estimator is $\hat{\pi} = \frac{n}{\sum_{i=1} Y_i}$.